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A mechanistic model for estimating ammonia volatilization from slurry applied to bare soil

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Abstract

The loss of ammonia by volatilization after slurry application may lead to large losses of soil–plant nitrogen. It is also a major source of atmospheric ammonia. These fluxes must therefore be accurately measured. However, volatilization depends on many features of the soil, climate and slurry, so that it is difficult to predict, or even to interpret and compare data from different experiments. We have developed a mechanistic model that simulates the influence of the various factors on volatilization, accounting for the transfers and equilibria in the topsoil and between the soil and the atmosphere. The model uses readily available input data, including soil, meteorological and slurry data. It includes energy balance and advection submodels, which make it suitable for field scale applications using simple meteorological data. Sensitivity analysis showed that soil pH has a large influence on volatilization. The model is also sensitive to soil adsorption capacity and some hydraulic characteristics (saturation water conductivity, water content at field capacity). It has been calibrated under real field conditions using experimental data for ammonia fluxes measured over two weeks after slurry application, with a micrometeorological method giving data at 15-min intervals. The soil, climate and slurry factors were also measured. The model provides a fair picture of the ammonia fluxes throughout the volatilization, including the total ammonia volatilized, the decrease in daily loss and their short-term (< 1 h) variations due to the influence of meteorological conditions on soil surface temperature and atmospheric diffusion. Lastly the model has been used to simulate the influence of various meteorological conditions and agricultural techniques on ammonia volatilization. This model will make it easier to interpret data from different experiments and will help to improve the emission submodel of atmospheric ammonia deposition models.
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Keywords: Slurry; Soil; Climate; Mechanistic model; Atmospheric ammonia

1. Introduction

The volatilization of ammonia from applied fertilizer ammoniacal nitrogen is known to be a major cause of soil–plant nitrogen (N) loss. It is also one of the largest sources of atmospheric ammonia (E.C.E.T.O.C., 1994). Most of this ammonia is deposited on plants and soils not far from its source (Asman and van Jaarsveld, 1992; Sutton et al., 1995). It is one of the main contributors to soil acidification and water eutrophication. Consequently, it is important to have a clear understanding of the volatilization process in order to develop reliable methods for reducing ammonia volatilization.

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There have been many experimental studies on ammonia volatilization and N balance using wind-tunnels, or micrometeorological measurements in the field (see e.g., Denmead et al., 1977; Lockyer, 1984; Hargrove, 1988; Jarvis and Pain, 1990; Sommer et al., 1991; Sommer and Olesen, 1991; Moal et al., 1995; Katz, 1996). But integrating these findings is difficult, because volatilization depends on many factors, and experimental conditions vary greatly: ammonia losses are affected by the nature of the slurry, soil conditions, agricultural practices and climatic conditions (Ernst and Massey, 1960; Faurie and Bardin, 1979a,b; Terman, 1979). It is also difficult to estimate ammonia losses, and the result may depend on how the measurements are made (Hargrove and Kissel, 1979; Denmead and Raupach, 1993). Consequently, accurate, reliable data obtained under field conditions are lacking (Hargrove, 1988).

Practical tools are also required to better understand the influence of soil, climatic and technical factors on ammonia volatilization, and hence find different ways to reduce ammonia emissions from agriculture. The difficulties in estimating emissions suggest that modelling may help make understanding and estimating the influence of the different factors on volatilization more accessible, and make comparing the experimental results obtained under different conditions a practical possibility.

Many empirical models have been derived from experiments run under semi-controlled conditions (see Sommer et al., 1991; Sommer and Olesen, 1991; Moal et al., 1995; Katz, 1996), but each model only includes the influence of a limited number of factors, and they can only be used for predicting losses under conditions similar to those under which they were established. In contrast, mechanistic modelling tries to describe the volatilization process itself by simulating nitrogen transfers, transformations and equilibria after slurry application in the field. Rachhpahl-Singh and Nye (1986a,b,c, 1988) developed a mechanistic model simulating the simultaneous transformation and movement of urea and ammoniacal N in the soil, and the process of ammonia volatilization from the soil surface. It was based on conservation equations for urea, ammoniacal N and pH. Numerical solutions were obtained for specific initial and boundary conditions. The soil compartment of this model was well described, but water transfers were not taken into account. Another limitation was that the parameters describing the transfer of ammonia to the atmosphere were only defined for laboratory experiments. Van der Molen et al. (1990) developed the first mechanistic model for ammonia volatilization after the application of cattle slurry in the field. The transfers within the soil and to the atmosphere were better described than in the model of Rachhpahl-Singh and Nye (1986a,b,c). This model was used to simulate ammonia volatilization over plots of only several hundreds of square meters in field conditions. Mechanistic models explicitly account for the influence of most soil, climatic and slurry factors on ammonia volatilization (Sherlock and Goh, 1985a,b; Rachhpahl-Singh and Nye, 1986a,b; Kirk and Nye, 1991a,b; Van der Molen et al., 1990), but they usually need a large number of parameters and they require many input data which are difficult to obtain under field conditions. The model of Rachhpahl-Singh and Nye (1986a,b,c) needs very specific information on soil and can be used only under laboratory conditions. The model of Van der Molen et al. (1990) is only suitable for small plots and needs a large number of parameters (eight) or input data (seven), some of which must be measured specifically. Lastly, slurry infiltration must be calibrated before each use. Thus, none of these models has yet been fully validated under field conditions.

The volatilization model proposed in this article is derived from the two above models. Our objective was to set up a model that gives a realistic description of all the implied mechanisms so that it can be used for research applications and under a wide range of environmental conditions. As it is designed to assess the influence of climatic data and agricultural techniques on volatilization, it should be applicable under field conditions. It must use readily-available input data and parameters, like data of a standard weather station, soil texture and standard slurry analysis.

2. Model description

2.1. Main features

The model deals with the chemical and physical equilibria between the various species of ammonia-

cal N in the soil, the transfer of heat, water and ammoniacal N within the soil, and the transfer of ammonia, heat and evaporation between the topsoil and the lower atmosphere. These processes are simulated with short time intervals over several days, or several weeks following a slurry application in the field. The main components of the model, including the compartments and their inter-relations are shown in Fig. 1. The model is composed of six submodels. Three of them deal with ammoniacal N transfers and equilibria between ammoniacal N species:

1. physical and chemical equilibria in the soil
2. aqueous and gaseous ammoniacal N transfers through the soil
3. gaseous ammonia transfer from the soil to the atmosphere.

The other three simulate heat and water transfer in the soil

4. water transfer in the soil
5. heat transfer in the soil
6. energy budget, water and heat exchange between the soil and the atmosphere.

The last three submodels are necessary because ammonia is transported with water in the soil, and the equilibria depend on the temperature and concentration in the soil water. Submodels (1), (2), (3) and (4) are the basis of the model: research applications may use a reduced model with only these four submodels. Submodels (5) and (6) make the model operational, as they allow calculation of several micrometeorological data, such as soil surface temperature and evaporation, from readily available meteorological data and surface descriptions.

The physical and chemical properties within each soil layer are considered to be uniform. The physical properties of the soil do not change with time after the slurry has been spread. We assume that urea was completely hydrolysed at slurry application, as urea is converted to ammoniacal nitrogen and carbonate within a few hours of excretion (Pakrou and Dillon, 1995). As ammonia volatilization is simulated over a relatively short time (2–3 weeks), nitrogen transformations by organic matter mineralization, ammoniacal nitrogen organization or uptake by plants, oxidation or nitrification are not accounted for. The mineralization of the organic N from the slurry is also considered to be negligible over the volatilization period.

Time and space have been discretized so as to provide a good compromise between computing time

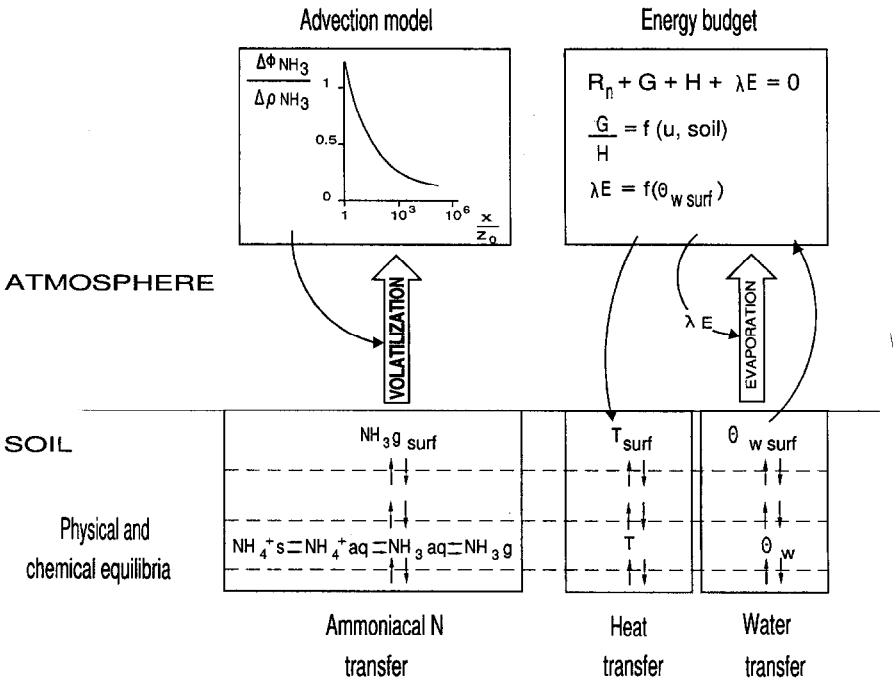


Fig. 1. Diagram of the volatilization model.

and model precision. The upper layer of the soil is divided into several layers, with limits between layers increasing quasi-geometrically: 2, 5, 10, 20, 50 and 100 cm. They can be changed to any depth progression. The time scale of each module is adapted to the mechanism described. It must be large enough to allow the physical and chemical equilibria to be established, but short enough to describe the rapid changes in surface fluxes, surface temperature and water transfer in the soil. The compromise between these constraints has resulted in a time step of 15 min.

2.2. Physical and chemical equilibria in each soil layer

To ensure simplicity, several assumptions about chemical equilibria are needed. Solutions are considered to be ideal aqueous solutions, with solute concentrations close to zero and solute activities equal to their molar concentrations. We thus assume that the theoretical values for ideal aqueous solutions can be used to describe the equilibria for slurry, although it is a complex solution of various ionic species. The effect of the ionic strength is not taken into account, although it has been shown that it can reduce ammonium ions activity to 75% of their concentration for some solutions analogous to slurries (English et al., 1980). Following Rachhpahl-Singh and Nye (1986a,b,c), we consider that the equilibria between the various forms of ammoniacal nitrogen are instantaneous.

The solubility of volatile weak electrolytes like ammonia results from two equilibria: ionization (Eq. (1)) and liquid–gas equilibria (Eq. (2)).

$$NH_4^+\ (aq) + H_2O \leftrightarrows NH_3\ (aq) + H_3O^+\ (aq) \quad (1)$$

$$NH_3\ (aq) \leftrightarrows NH_3\ (g) \quad (2)$$

where NH_4^+ (aq) is aqueous ammonium ions, NH_3 (aq) is aqueous ammonia and NH_3 (g) is gaseous ammonia. The proportions of NH_4^+ (aq) and of NH_3 (aq) are determined by the pH of the solution and the equilibrium constant, K_A, expressed as an exponential function of the temperature of the solution (Beutier and Renon, 1978) (Table 1):

$$K_A = [H_3O^+\ (aq)]\frac{[NH_3\ (aq)]}{[NH_4^+\ (aq)]} \quad (3)$$

where [] denotes the molar concentration of the solute (mol l^{-1}). The gas–liquid equilibrium is accounted for by the Henry's law constant K_H, also expressed as an exponential function of the temperature of the solution (Beutier and Renon, 1978) (Table 1):

$$P_{NH_3} = K_H[NH_3\ (aq)] \quad (4)$$

Combining Eqs. (3) and (4) provides the solubility expression as a function of the total aqueous ammoniacal N concentration of the soil solution ([AN (aq)]):

$$P_{NH_3} = K_H\frac{[AN\ (aq)]}{1 + \dfrac{[H_3O^+\ (aq)]}{K_A}} \quad (5)$$

A third equilibrium develops in the soil as ammonium ions are adsorbed by clay minerals and soil organic matter (NH_4^+ (s)) (Nommik and Vahtras, 1982):

$$NH_4^+\ (aq) \leftrightarrows NH_4^+\ (s) \quad (6)$$

Binding is assumed not to occur within the time interval of several days to a few weeks. The partitioning of ammonium ions between the soil solid fraction (NH_4^+ (s)) and the soil aqueous phase (NH_4^+ (aq)) is described by a Freundlich isotherm (Bolt, 1976):

$$[NH_4^+\ (s)] = a_s[NH_4^+\ (aq)]^{b_s} \quad (7)$$

2.3. Water transfers between the soil layers

The flux density equation for vertical flow of water in the soil is described by Darcy's law, generalized to the unsaturated zone:

$$J_W(\theta_W) = -K_W(\theta_W)\frac{\partial h_W(\theta_W)}{\partial z} + K_W(\theta_W) \quad (8)$$

where h_W is the pressure head, K_W the hydraulic conductivity and θ_W the water content of the soil. The soil is considered to be a homogenous, incompressible and isotropic porous medium, and the transport of liquid water to be vertical and one-dimensional. As the water content after slurry spread-

Table 1
Model equations and parameters

Reference	Parameter/definition
Acid–base equilibrium constant K_A (Beutier and Renon, 1978)	$\ln(K_A) = -177.95 - \frac{1843}{T} + 31.434\ln(T) - 0.0545T$
Henry's law constant K_H (Beutier and Renon, 1978)	$\ln(K_H) = 160.46 - \frac{8621}{T} + 25.677\ln(T) + 0.0354T$
Pedo-transfer functions (Clapp and Hornberger, 1978)	$h_W(\theta_W) = h_{Wsat}\{(\theta_{Wsat})/(\theta_W)\}^{bw}$ and $K_W(\theta_W) = K_{Wsat}\{(\theta_W)/(\theta_{Wsat})\}^{2b_W+3}$
Thermal conductivity (Noilhan and Planton, 1989)	if $\ln(h_W(\theta_W)) > 5.1$ then $\lambda_S = 0.171$; if $\ln(h_W(\theta_W)) < 5.1$ then $\lambda_S = \exp(-\ln(h_W(\theta_W)) + 2.7)$
Diffusion coefficient of aqueous ammoniacal N in the soil liquid phase (Millington–Quirk, in Jury et al., 1983)	$D^W_{NH_4^+} = \{(\theta_W^{10/3}/(\phi^2)\}D^{aq}_{NH_4^+}$
Theoretical diffusion coefficient of aqueous ammoniacal N in free water (Robinson and Stokes, 1959)	$D^{aq}_{NH_4^+} = (RT\lambda_{NH_4^+})/(\lvert Z_{NH_4^+}\rvert F^2)$
With the limiting equivalent conductance	$\lambda_{NH_4^+} = 10.56 + 90.72 + \log(Z_{NH_4^+}) + 42.95\{(\gamma_{NH_4^+})/(Z_{NH_4^+})\}$
Empirical diffusion coefficient of aqueous ammoniacal N in free water (Van der Molen et al., 1990, from data of Yuan-Hui and Gregory, 1974)	$D^{aq}_{NH_4^+} = (9.8\times10^{-10})1.03^{(T-273)}$
Diffusion coefficient of aqueous ammoniacal N in the liquid phase of the soil (Millington–Quirk model, in Jury et al., 1983)	$D^g_{NH_3} = \{([\phi - \theta_W]^{10/3})/(\phi^2)\}D^{air}_{NH_3}$
Empirical diffusion coefficient of aqueous ammoniacal N in free water (Van der Molen et al., 1990, from data of Yuan-Hui and Gregory, 1974)	$D^{air}_{NH_3} = (1.7\times10^{-5})1.03^{(T-293)}$
Coefficients a_i and b_i for the analytical solution of local advection due to a step change in concentration(Itier and Perrier, 1976)	for $10 \le \frac{x}{z_0} \le 10^3$, $a_i = 0.887$ and $b_i = 0.19$; for $10^3 \le \frac{x}{z_0} \le 10^5$, $a_i = 0.665$ and $b_i = 0.15$

ing is generally high, water vapor transfers are considered to be negligible. For the same reason, convective transfers dominate, and water and heat fluxes are not coupled (Ten Berge, 1990). Soil hydraulic properties are estimated using the Clapp and Hornberger (1978) pedo-transfer functions, which relate the pressure head h_W and the hydraulic conductivity K_W to the soil water content θ_W, by considering only the soil texture (Table 1).

The sudden supply of water at slurry application causes soil compaction, which reduces soil surface infiltrability. It is followed by the ponding of slurry at the soil surface. This effect is simulated by multiplying the hydraulic conductivity of the surface layer by an infiltrability coefficient (Fiès and Panini, 1995).

2.4. Heat transfers between the soil layers

Heat transfers in the soil and temperatures of each soil layers are calculated following the equation:

$$C\frac{\partial T}{\partial t} = \frac{\partial}{\partial z}\left(\lambda_S\left(\frac{\partial T}{\partial z}\right)\right) \tag{9}$$

The thermal conductivity (λ_S) uses the parameters of Noilhan and Planton (1989) (Table 1) and the volumetric heat capacity (C) uses the parameters of de Vries (1963). Interactions between heat and water flow are not taken into account.

2.5. Aqueous and gaseous ammoniacal N transfers between the soil layers

From Fick's law, the transfer of aqueous solute in a porous medium is described by the classical convection–diffusion scheme (Bear, 1972):

$$\frac{\partial([\mathrm{AN\ (aq)}]\theta_W)}{\partial t} = -\frac{\partial(J_W(\theta_w)[\mathrm{AN\ (aq)}])}{\partial z} + \frac{\partial}{\partial z}\left(D^W_{NH_4^+}\frac{\partial[\mathrm{AN\ (aq)}]\theta_W}{\partial z}\right) \tag{10}$$

The ammoniacal N diffusion coefficient in the soil ($D^W_{NH_4^+}$) depends on the ammoniacal N diffusion coefficient in water, and on the soil water content and porosity, integrated in a tortuosity factor (model of Millington–Quirk, in Jury et al., 1983) (see Table 1). The diffusion coefficient of ammoniacal N in water is taken to be the same for ammonia and ammonium ions. It depends on the solute species and on the temperature (Robinson and Stokes, 1959) (Table 1). The parameters used by Van der Molen et al. (1990) from tabulated values of Yuan-Hui and Gregory (1974) are chosen (Table 1).

The ammonia gas transfer equation is analogous to the diffusive part of the aqueous ammoniacal N transfer in the soil:

$$\frac{\partial([\mathrm{AN\ (g)}](\phi-\theta_W))}{\partial t} = \frac{\partial}{\partial z}\left(D^g_{NH_3}\frac{\partial[\mathrm{AN\ (g)}](\phi-\theta_W)}{\partial z}\right) \tag{11}$$

The diffusion coefficient in the soil atmosphere ($D^g_{NH_3}$) is also derived from the model of Millington–Quirk (Jury et al., 1983) (Table 1). The diffusion coefficient in free air was derived from the work of Bruckler et al. (1989) by Van der Molen et al. (1990) (Table 1).

2.6. Transfer of ammonia gas to the atmosphere

Itier and Perrier (1976) developed a model of local advection and showed that change in the surface flux, $\Delta\Phi$, is expressed as a function of the distance from the leading edge in the wind direction, x, and the difference in surface concentration, $\Delta\rho$, for an abrupt change in surface concentration with no roughness change.

$$\frac{\Delta\Phi(x)}{ku^*\Delta\rho(x)} \approx a_i\left(\frac{x}{z_0}\right)^{-b_i} \tag{12}$$

where z_0 is the roughness length of the field, depending on the soil surface geometry. Its value can be calculated or obtained for some surfaces from tabulated values (Brutsaert, 1982; Guyot, 1997). Coefficients a_i and b_i vary with the ratio x/z_0 (Itier and Perrier, 1976) (Table 1). This theory can apply to the volatilization of ammonia after slurry application, with a step change in ammonia surface concentration. The background ammonia concentrations and fluxes over surrounding fields where no nitrogen had been applied for the previous days, are commonly assumed to be zero. For a field with length in the wind direction X, Itier and Perrier (1976) also showed that the mean value of the flux over the field

is equal to the flux measured or calculated at a distance which does not depend on the ratio x/z_0, $0.3X$ downwind from the edge of the field. The mean ammonia surface flux of the field can thus be expressed as:

$$\Phi_{NH_3} = a_i(ku^*)\rho_{NH_3}\left(\frac{0.3X}{z_0}\right)^{-b_i} \quad (13)$$

2.7. Energy budget, water and heat exchange between the soil and the atmosphere

Soil surface temperature and evaporation are calculated using the energy balance equation of the bare soil surface:

$$R_n - G = H + \lambda E \quad (14)$$

where R_n is the net radiation, G is the soil heat flux by conduction, H is the sensible heat flux to the atmosphere and λE is the soil evaporation. All fluxes are expressed in W m^{-2}. Over the time interval of 15 min, each term of this equation and the equilibrium term, surface temperature, are constant. Their parameters are set so that they can be calculated from standard meteorological data and readily available soil parameters.

Net radiation can be described as:

$$R_n = (1-a)R_s + \varepsilon_s(R_a - \sigma T_s^4) \quad (15)$$

where R_s is the measured solar radiation. The soil emissivity, ε_s, is taken as 1. The soil albedo a can be measured or obtained from tabulated values for bare soil (Guyot, 1997).

The atmospheric radiation, R_a is considered to be constant throughout the day, and is calculated from daily averages of air temperature (T_{ad}) and humidity (e_{ad}), following Brutsaert (1975, 1982):

$$R_a = \varepsilon_a \sigma T_{ad}^4 \quad (16)$$

where ε_a is the atmospheric emissivity:

$$\varepsilon_a = \varepsilon_{aB} + n(1-\varepsilon_{aB})\left(1 - 4\frac{11}{T_a}\right) \quad (17)$$

with

$$\varepsilon_{aB} = 1.24(e_{ad}/T_{ad})^{1/7} \quad (18)$$

The cloudiness, n, can be estimated from the ratio of measured daily solar radiation to the extraterrestrial radiation, calculated from astronomical formulae (Durand and Legros, 1981; Spitters et al., 1985).

The sensible heat flux, H, is proportional to the temperature difference between the soil surface and a reference height in the atmosphere. The proportionality coefficient, r_a, is expressed as a resistance including stability corrections, according to the theory of turbulent transfer in the surface boundary layer (Brutsaert, 1982):

$$H = \rho c_p (T_s - T_a)/r_a \quad (19)$$

Soil evaporation is similarly defined, using the difference in air humidity:

$$\lambda E = \rho c_p \gamma [hP(T_s) - P(T_d)] r_a \quad (20)$$

where $P(T)$ is the saturation vapour pressure at temperature T, and h is a relative humidity in the top soil layer calculated following Noilhan and Planton (1989):

$$h = 0.5[1 - \cos(\pi\theta_{Wsurf}/\theta_{Wfc})] \quad (21)$$

where θ_{Wsurf} is the top soil water content calculated from the soil water transfer model, and θ_{Wfc} is soil water content at field capacity.

The soil heat flux, G, is estimated from its ratio to H with a simple single function of the daily average of wind speed, u_d, for all soil conditions, including a phase shift, φ_{GH}, between these two fluxes (Cellier et al., 1996):

$$G/H = \varphi_{GH} 1.36/u_d^{1/2} \quad (22)$$

2.8. Calculations

The exchange of ammonium ions between the soil solid and aqueous phases was simulated only in the first soil layer, where the total ammoniacal nitrogen content (TAN) is large. The partitioning of ammoniacal N between the various forms and phases was obtained by solving Eqs. (5) and (7) using an iterative procedure using a time interval of 15 min. The TAN was taken as the initial value for aqueous ammoniacal N content in the soil (AN aq). Iterations were performed until the difference in aqueous ammoniacal N content between two successive iterations was less than 0.1% of TAN, or when the aqueous ammoniacal N content of the layer was less than 10^{-5} mol m^{-2}.

Eqs. (14)–(22) were also solved for each time step of the model using an iterative procedure. The initial value of the soil surface temperature was taken to be the air temperature. The iterative procedure was completed when the difference between two successively calculated surface temperatures was less than 0.01°C. The surface temperature and the evaporation flux were then used in the heat- and water-transfer submodels.

The transfers of heat (Eq. (9)), water (Eq. (8)) and solute (Eqs. (10) and (11)) within the soil were calculated using a basic time interval of 15 s, for the sake of numerical stability, depending on the depth of the soil layer (Wagenet, 1981). For the first soil layer, rainfall, evaporation rates and ammonia volatilization obtained for a time interval of 15 min were broken down for a time interval of 15 s. The water and total ammoniacal N supplied by the slurry were assumed to be added to the first soil layer at the time of slurry application.

2.9. Summary

Most of the processes involved in ammonia volatilization are described with transfer or equilibrium equations. Their coefficients were chosen independently of the validation experiments. Table 2 sums up the entry data and variables, and model parameters, and their availability, or the way they are obtained. Although it is a mechanistic model, entry data and variables and its parameters are rather easy to obtain. However, some specific measurements are needed.

The coefficients of the Freundlich isotherm describing ammonium exchange between the soil aqueous and solid phases are not usually determined. It is difficult to predict the soil ammonium exchange capacity and there is no standard procedure for its determination, but even if it were available it would be difficult to apply to the soil in field conditions the properties of an oversaturated and disaggregated soil measured in the laboratory. As a matter of fact, in the field, soil water content is mostly at field capacity or below, and the soil is aggregated.

The change in the pH of the soil–slurry complex is not taken into account and is not described in the model, although pH is recognized to be an important factor determining ammonia volatilization. Its evolution is linked to ammonia volatilization: the reactions shown in Eq. (1) and Eq. (2) show that one proton is produced for each molecule of NH_3 volatilized. The change in soil–slurry pH is also linked to the presence of buffering substances that can neutralize this proton, especially bicarbonate ions, which are the main proton acceptors in the soil–slurry solution (Fordham and Schwertmann, 1977). Few investigations have been conducted on the buffering capacity of slurries (Sommer and Husted, 1995a,b) or soils (Pierre, 1927), for the same reasons as there have been few studies on the ammonium exchange capacity. No mechanistic nor empirical information is yet available, so that the changes in pH during ammonia volatilization can not yet be readily simulated. Consequently, the ammonium retention capacity and soil–slurry pH must be calibrated.

3. Model testing

The model was designed to simulate ammonia volatilization from any field where slurry has been applied by a farmer using standard application practices. It must therefore be tested under real agricultural conditions. This was done by comparing calculated ammonia fluxes with observed fluxes measured during volatilization after spreading slurry over a bare soil. Ammonia fluxes and the various factors influencing ammonia volatilization were monitored over a two-week period after slurry application.

3.1. Materials and methods

3.1.1. Field conditions

The experiment was carried out at the INRA experimental station of Le Rheu, a few kilometers west of Rennes (Brittany, France). The slurry was applied on 16 March 1994 (DOY (= day of year) 75) from 0820 h (UT) to 1600 h (UT) over a 1.7-ha (90×190 m) field. The soil was a slightly acid (pH = 6.6) loam with 13.4% coarse sand and gravel. It was a highly crusting soil with a porosity of 0.53 $m^3\ m^{-3}$ and a cation exchange capacity (CEC) of 6.5 meq $(100\ g\ soil)^{-1}$. The previous crop was maize, whose crushed residues were left on the soil surface. The slurry was from dairy cattle and had a low dry matter content (4.7%), a low ammoniacal N

Table 2
Input data, input variables, and parameters

Input data, input variables, or parameters	Availability of source	Submodel where it is used
slurry		
ammoniacal N content (TAN)	slurry analysis	physical and chemical equilibria
dry matter content	gravimetry	water transfers
soil		
initial water content (θ_w)	gravimetry	all except energy budget
texture	soil analysis	water, heat transfers, energy budget
albedo (a)	measurement or tabulated values	energy budget
adsorption coefficient (a_s, b_s)	specific analysis	physical and chemical equilibria
soil–slurry complex		
pH change	direct measurement	physical and chemical equilibria
agricultural practices		
amount of slurry applied	count of number of tanks	water and ammoniacal N transfers, and physical and chemical equilibria
field length (X)	plans	ammonia volatilization
roughness length (z_0)	tabulated values	ammonia volatilization
climatic conditions		
solar radiation (R_S)	meteorological station	energy budget
air temperature (T_a)	meteorological station	energy budget
air humidity (T_d and e_a)	meteorological station	energy budget
wind speed (u_d)	meteorological station	energy budget, ammonia volatilization
rainfall	meteorological station	energy budget, water transfers
wind direction	meteorological station	ammonia volatilization

concentration (0.86 kg N m^{-3}) and an acid pH (7.1). The soil water content (31.3%) was near field capacity when the slurry was applied. Slurry was spread on the field from a tank at a rate of 133 m^3 ha^{-1}, for a total ammoniacal nitrogen supply of 114 kg N ha^{-1}.

3.1.2. Ammonia flux measurements

Micrometeorological methods were used because they are the only methods suitable for the continuous, direct measurement of ammonia fluxes at short time intervals. Also, they do not alter the overall microclimatic conditions which influence volatilization (Denmead and Raupach, 1993). Two independent micrometeorological methods were used to monitor ammonia fluxes (for experimental details, refer to Génermont et al. (1996, 1997)).

A mass balance method was used during the first few hours after spreading. This method equates the vertical flux from the edge of the field to the mast to the horizontal flux through a vertical plane located at the measurement mast (Denmead et al., 1977). Ammonia concentration and wind speed were measured at five heights (0.44, 0.80, 1.50, 2.50 and 5.00 m) above the soil surface, at time intervals of less than 1 h for the first few hours after spreading, and every 4 h at the end of the first 24 h. The ammonia concentrations were obtained by acid trapping in 50 ml 0.072 N H_2SO_4, with an air flow rate through the solution of about 3.5 l min^{-1}. The ammonium concentration was determined by colorimetric analysis (RFA 300, Perstcorp Analytical, Silver Spring, USA) in the laboratory.

A simplified aerodynamic method was used when the surface covered with slurry was large enough so that the constant flux layer was adequately deep. These measurements were continued for 15 days from the end of the day of spreading (1900 h UT). This method expresses the vertical flux in a diffusive form, the transfer coefficient depending on the wind speed and air temperature gradient calculated from two measurement heights (Itier, 1981). The wind speed and air temperature were measured at 0.5 and 1.5 m above the soil surface. The ammonia concentrations at the same heights were measured with a chemiluminescence analyser (NH30M, Environnement S.A., Poissy, France). The ammonia concentration at both heights were determined using the same analyser to ensure precision. Alternating samples from each height were pumped to the analyser through a 2 m long, 6 mm diameter Teflon tube, using a Teflon valve. Because of the time constant of the analyser, the valve was switched every 7.5 min, and the measured values were recorded 5 min after switching.

All air temperatures were measured using copper–constantan thermocouples (AWG 30, Thermoelectric, Limeil Brevannes, France) placed in ventilated shields, and wind speed was measured with cup anemometers (CE 155, Cimel, Paris, France).

3.1.3. Other measurements

Evaporation, λE, was estimated as the residual term of the energy balance (Eq. (22)). Net radiation, R_n, was measured using a net radiometer (type S1, Swissteco, Oberriet, Switzerland) placed 2 m above the soil surface. The soil heat flux, G, was measured with a fluxmeter (Heat Flow Transducer 3.1, Radiation and Energy Balance Systems, Seattle, WA, USA), and the energy storage above the location of the heat flow transducer was calculated using the calorimetric method with the soil temperature profile. The sensible heat flux to the atmosphere, H, was estimated by the simplified aerodynamic method (Itier, 1981) using the system described above for ammonia flux measurements.

Incident and reflected solar radiation flux densities were measured 1 m above the soil surface with two pyranometers (CM6, Kipp and Zonen, Delft, Netherlands), air relative humidity at 0.5 and 1.5 m with a hygrometer (HMP35A, Vaisala, Helsinki, Finland), rainfall with a tipping bucket rain gauge (Précimécanique, Paris, France) with 0.1 mm per tip. Soil surface temperature was measured using six copper–constantan thermocouples (AWG 24, Thermoelectric Limeil Brévannes, France) fixed on the soil surface with a thin plastic stem. Soil temperature was also measured at the depths 2, 5, 10, 20 and 50 cm using 4, 4, 3, 2 and 1 copper–constantan thermocouples (AWG 30, Thermoelectric Limeil Brévannes, France).

All micrometeorological data were recorded every 5 s and averaged over 15 min intervals with data loggers (Model CR10, Campbell Scientific, Shepshed, UK).

Soil water and ammoniacal nitrogen contents were monitored before slurry spreading and during volatilization on days 1, 2, 4, 6, 12 and 26 after spreading for layers 0–10 and 10–25 cm, with three replicates. Each sample was taken from nine well-mixed subsamples. Water content was measured by gravimetry. Ammoniacal N was extracted with KCl and measured using a colorimeter analyser (RFA 300, Perstcorp Analytical). More precise profiles were also constructed over the first 10 cm of soil, divided into five layers of 2 cm.

Soil surface pH was monitored every hour for the day of spreading and the first day after spreading, and three times a day for the following days, with one or two replicates: each sample was taken from 10 mixed subsamples. Twenty grams of the fresh soil–slurry complex were diluted in 250 ml pure water, and the pH was measured during mixing using a KCl electrode connected to an electronic pH-meter (C425, Consort, Turnhout, Belgium) after 15 min of mixing and 30 min of decanting in a closed small vessel.

3.2. Sensitivity analysis

The sensitivity of the model to the parameters of the ammoniacal N transfers in the soil and in the atmosphere, and of the physical and chemical equilibria in the soil were analysed. This analysis was performed using only the first four submodels to avoid introducing bias from the energy balance and soil heat transfer submodels, which are only useful for making the model operational (see Section 2.1 for an overall description of the model). The chosen parameters were those most influencing ammonia volatilization: (1) pH of the soil solution, (2) the equilibrium constants K_A and K_H and (3) the ionic strength (I) of the solution, for the chemical equilibrium in the soil; (4) the parameters of the soil hydraulic properties θ_{Wsat}, h_{Wsat}, K_{Wsat} and b_w for the transfers of water and ammoniacal N, and (5) soil surface roughness (z_0), and background concentration for the simulation of the gaseous ammonia transfer to the atmosphere. The imposed range of variation depended on the actual range of variation of the parameter but also on its determination uncertainty.

The criteria used to analyse the sensitivity was the ratio of the cumulative losses with a given value of a factor to the losses in the reference simulation. But the response of volatilization to anyone factor may be very different over short and long periods. For long periods, it depends on the direct effect of the factor, and on the availability of ammoniacal N. Consequently, the model sensitivity was tested over a short period (24 h) to describe the immediate response of the model to a change in one parameter, and over a long period (10 days) to evaluate its residual effect on the availability of ammoniacal N at the soil surface (Figs. 2 and 3a,bFig. 4a–d). The variations for most factors were large for the first day, but much less for the 10-day period. The effect of one factor on ammonia losses could even be reversed after the first day.

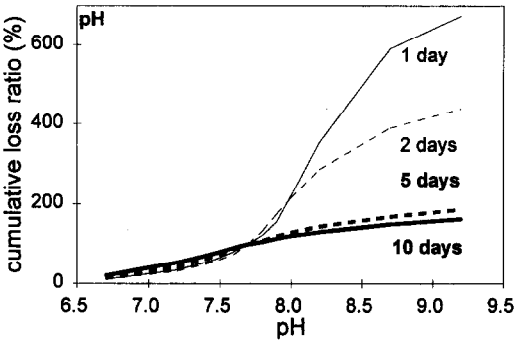


Fig. 2. Sensitivity of the model to pH. The cumulative loss for a given value of the studied factor is plotted against the loss of the reference simulation over a 24-h period (thin line), a 2-day period (thin broken line), a 5-day period (thick broken line) or a 10-day period (thick line).

The most sensitive factor was the pH (Fig. 2), especially when it increased. The increase in ammonia loss with increasing pH was due to driving the acid–base equilibrium (Eq. (1)) from the ammonium to the ammonia form. Aqueous ammonia was the main species in the soil solution at the temperatures measured during the experiment, and for a pH above 7.8, while ammonium was negligible. The pH determined not only the total loss, but also the pattern of ammonia emission. Fluxes were low at low pH, but there were emissions over the whole volatilization period; fluxes were very high for the first few days when the pH was high, but they decreased very rapidly and became negligible sooner than when the soil was acid.

The model was much less sensitive to the equilibrium constants than to the pH. It was low over 10

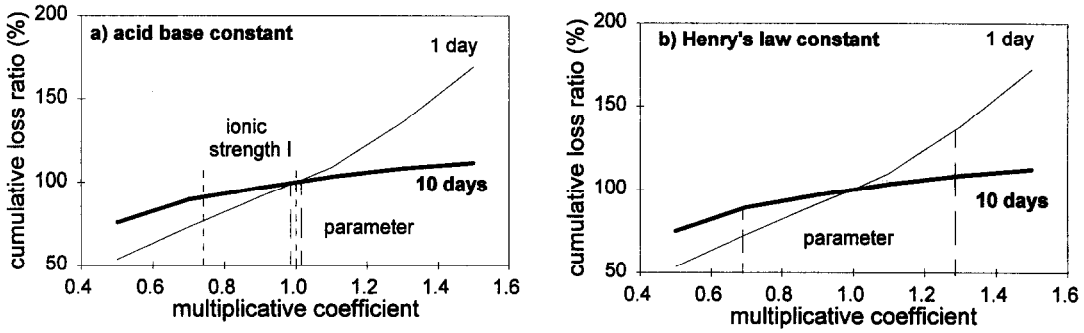


Fig. 3. Sensitivity of the model to the equilibrium constants K_A (a) and K_H (b). The cumulative loss for a given value of the studied factor is plotted against the loss of the reference simulation over a 24-h period (thin line) or a 10-day period (thick line).

days (Fig. 3a,b). The influence of the acid–base equilibrium constant on ammonia loss was similar to that of the Henry's law constant but the parameters of the Henry's law constant differ by over 25% between authors (Beutier and Renon, 1978; Hales and Drewes, 1979), while acid–base constant differs by less than 2% (Bates and Pinching, 1950; Beutier and Renon, 1978). The uncertainty of error of the global equilibria (Eqs. (1) and (2)) (less than 30% over 1 day and less than 10% over 10 days) was then mainly due to the liquid–gas equilibrium. Not taking into account the ionic strength resulted in overestimating the 24-h fluxes by 25%, and the 10-day losses by only 10% (Fig. 3a,b). Thus, it is not

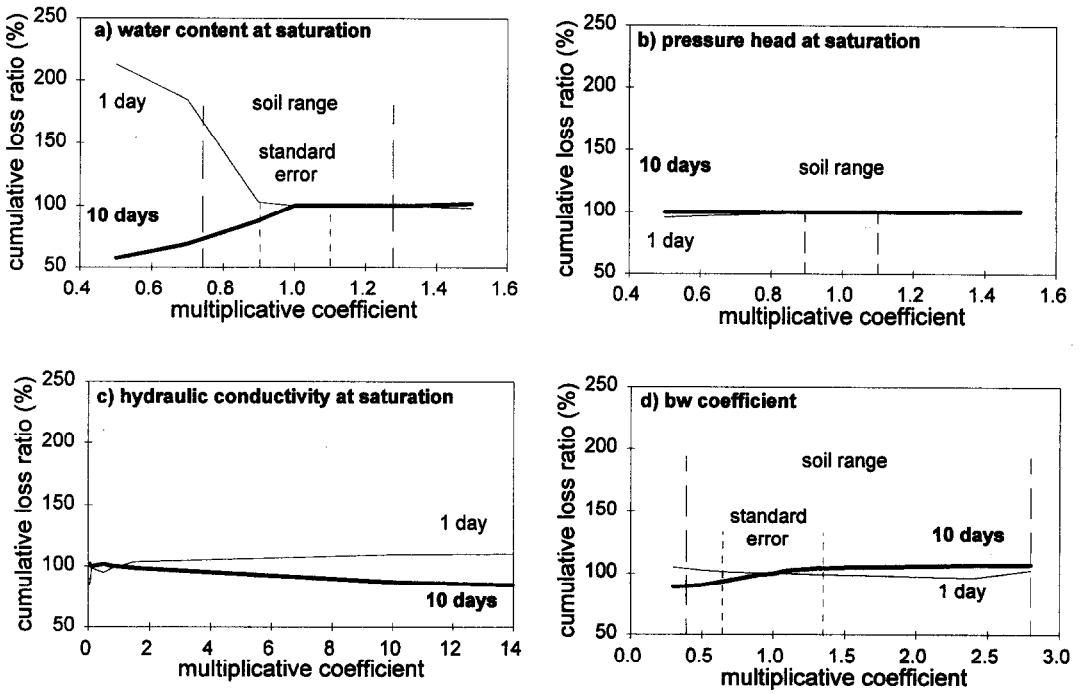


Fig. 4. Sensitivity of the model to the parameters of the soil hydraulic properties. The cumulative loss for a given value of the studied factor is plotted against the loss of the reference simulation over a 24-h period (thin line) or a 10-day period (thick line). (a) soil water content at saturation: θ_{Wsat}; (b) soil water pressure head at saturation: h_{Wsat}; (c) hydraulic conductivity of the soil at saturation: K_{Wsat}; (d) coefficient of the pedo-transfer function: b_W.

absolutely necessary to include the effect of all the other ions on the chemical equilibria even in a very complex aqueous solution like a slurry.

The accuracy of the water and ammoniacal N transfer descriptions was largely due to the accuracy of the parameters of the soil hydraulic properties. The model was sensitive to the saturation water content (θ_{Wsat}) (Fig. 4a). Underestimating θ_{Wsat} underestimated the porosity, decreasing water transfers in the soil, and leading to an overestimate of ammonia fluxes for the first day because ammonia remained at the soil surface. The reverse response over 10 days might result from complex interactions between θ_{Wsat}, hydraulic conductivity and head pressure, which were more significant over long periods. However, this parameter was generally fairly well estimated with standard errors of less than 15% (Clapp and Hornberger, 1978). The effect was then small. The model was slightly sensitive to the saturation pressure potential (h_{Wsat}) (Fig. 4b): less than 6% for one day, and less than 1% over 10 days. As a consequence, the great uncertainty in estimating the saturation pressure potential, with standard errors of more than 100% (Clapp and Hornberger, 1978), would not affect the accuracy of the model. The model was also only slightly sensitive to the saturation hydraulic conductivity (K_{Wsat}) and to the b_w coefficient (Fig. 4c and d). But the effect of the b_w coefficient was greater at 10 days (11%) than over one day (6%), although it was reverted. This coefficient acted directly on the water content at field capacity of the soil. We conclude that, though the description of the hydraulic properties of the soil using pedo-transfer functions gave only rough estimates of their values, it was sufficient for this application.

For the advection submodel, the main parameter influencing the ammonia volatilization flux was the roughness length. It had a limited effect on ammonia losses over one or 10 days, in the range of variations for bare soils (Guyot, 1997). The assumption that the background concentration of ammonia gas was equal to zero has few consequences for the results, because increasing the background concentrations to 50 $\mu g\ m^{-3}$ changed the response of the model by less than 3%. As a whole, the model was not sensitive to the parameters of the advection submodel.

The volatilization model could be calibrated using the most sensitive factors for the submodels, for which there was an uncertainty in the measure or in the estimate. The most sensitive factor for the physical and chemical equilibria was pH, and a great uncertainty was pointed out for this factor. The soil adsorption capacity was also pointed out for the uncertainty in its estimate. K_W was the most sensitive parameter for water transfers, and b_W seemed to have a particular effect on water and ammoniacal N transfers. The other submodels did not need calibration. They have been widely validated for other experimental data (Itier and Perrier, 1976; Cellier et al., 1996).

3.3. Calibration and comparison with field measurements

3.3.1. Data used

The initial measured water content was very close to the soil water content at field capacity (31.5%). Following Guyot (1997), soil surface roughness was taken to be 0.01 m and soil albedo as 0.2. The adsorption isotherm coefficients were not measured because we lacked a standardized protocol. The only published parameters available, for a sandy-loam soil were used (Rachhpahl-Singh and Nye, 1984):

$$\left(NH_4^+\ (s)\right) = 4\left(NH_4^+\ (aq)\right)^{0.602} \qquad (23)$$

The pH was not simulated by the model. We used a constant pH over the whole volatilization period, or the data measured, and any missing values were graphically interpolated to obtain data for 15-min intervals.

Three independent calibrations, water transfers, ammoniacal N transfers and physical and chemical equilibria, were done for the complete model. The water transfer submodel was calibrated first by reducing the infiltrability of the first soil layer to 2% of its initial value, so that slurry infiltration on the first day was slowed. This value was close to values estimated for highly crusting soils, as in the present case. Observations over a wide range of soils showed that it is usually 1–10% (Fiès and Panini, 1995). However, water still infiltrated too fast during the first day (Fig. 5a). The calibration was also necessary because the simulated soil water content after slurry had infiltrated the first soil layers was lower

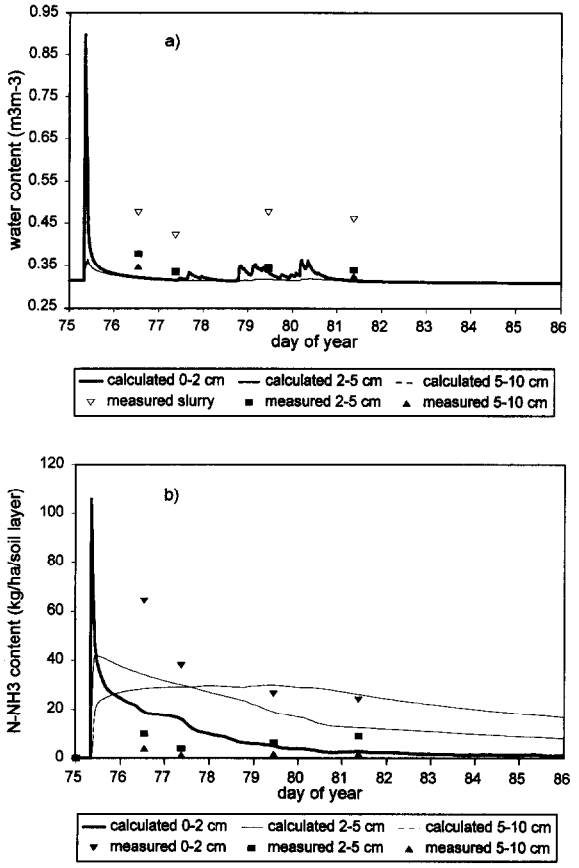


Fig. 5. Measured and calculated soil water content (a) and ammoniacal N profiles (b).

than the measured content. Thus, the soil water at field capacity was significantly underestimated by the Clapp and Hornberger (1978) parameters. The water content after slurry infiltration was adjusted by increasing b_W for the first three layers of the soil. These increases (20, 19 and 1%) remained within the standard error range for this coefficient (standard error = 40%) (Clapp and Hornberger, 1978).

Because the slurry infiltrated into the soil too fast, ammoniacal N transfer was also too fast. The availability of ammoniacal N at the soil surface and ammonia fluxes were then underestimated. As it was impossible to actually estimate the ammonium adsorbed by the soil solid phase, the physical and chemical equilibria submodel was calibrated by adjusting the adsorption of ammonium in the first layer of the soil. The best fit of simulated ammoniacal N profiles to measured profiles was obtained with a_s = 0.3 and b_s = 0.6 (Fig. 5b). However, the transfer of ammoniacal N immediately after spreading remained too fast because of the water infiltration.

The last calibration was to adjust the physical and chemical equilibria submodel. The sensitivity analysis showed that pH was the factor that influenced ammonia volatilization most. We therefore selected a pH value that remained constant over the whole volatilization period and fitting ammonia emissions. The initial measured soil pH of 7.5 underestimated the first day fluxes, and the total losses. A constant pH of 7.8 for the whole ammonia volatilization period best fitted the total losses and the first day fluxes (Fig. 6a).

4. Discussion

This model takes into account the main physical and chemical mechanisms involved in the volatilization of ammonia following slurry spreading. The physical and chemical equilibria in the soil, and transfers of ammoniacal N within the soil and to the atmosphere are all described using a mechanistic approach, with similar degrees of accuracy, using input data that can be collected reasonably easily. It thus accounts realistically for the influence of slurry properties, soil conditions, climatic conditions and agricultural practices on each process affecting ammonia volatilization. The model is therefore adaptable to any system emitting ammonia after the application of ammoniacal nitrogen.

4.1. Comparison of simulated and measured volatilization

The simulation was satisfactory for the global amount because of the calibration procedures: the calculated loss was 57 kg ha^{-1} and the measured loss was 62 kg ha^{-1}. The fluxes for the first day were underestimated. As a consequence, hourly fluxes were slightly overestimated for DOY 78 to 83, because more ammoniacal N was available at the soil surface for ammonia volatilization (Fig. 6a). However, ammoniacal N was still transferred too fast during the first few hours. The daily pattern of ammonia emissions over a 11 day period was fairly well simulated: the calculated fluxes gradually de-

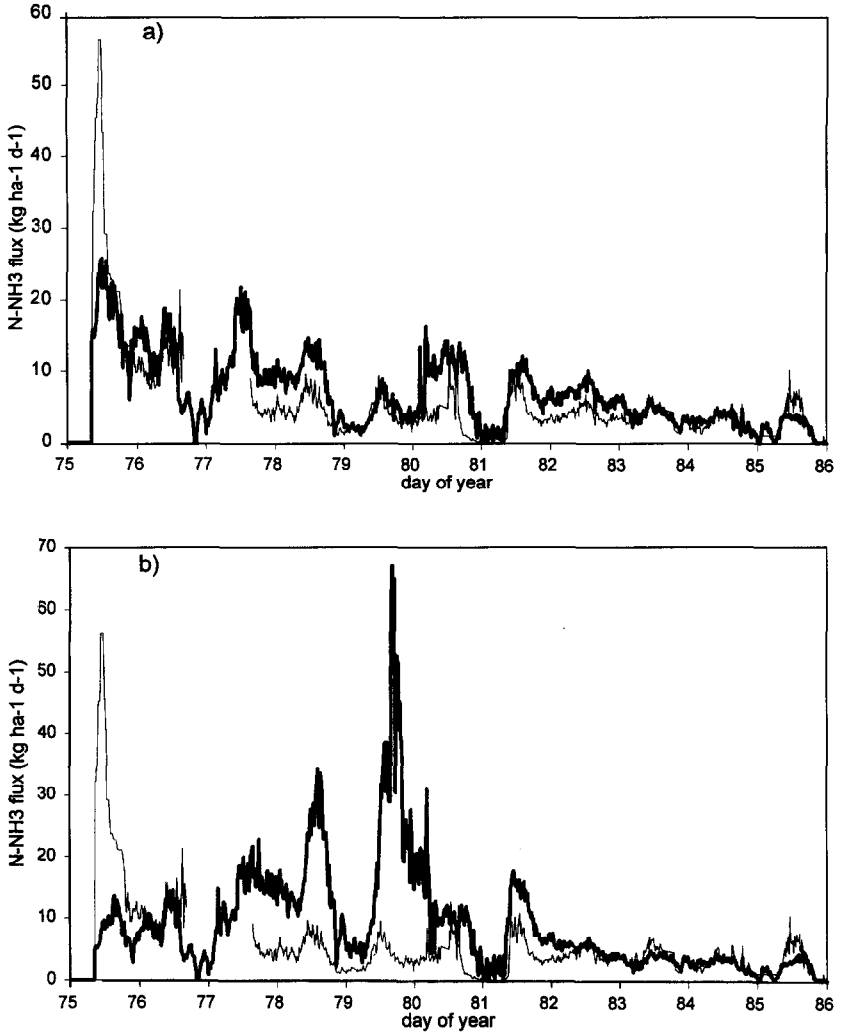


Fig. 6. Hourly measured (thin line) and calculated (thick line) ammonia fluxes at a constant pH = 7.8 (a) and with the pH varying over the volatilization period (b).

creased owing to the decrease in available ammoniacal N at the soil surface. The renewals of emissions due to the increases in solar radiation and in surface temperature were well described for DOY 80 and DOY 81, but not for DOY 85. The typical diurnal pattern (large fluxes during the day and small ones during the night) were well described, with maximum and minimum values being well estimated for most days. The response of fluxes to hourly variations in meteorological conditions were also well described, except for DOY 80. This showed that the advection and the energy balance submodels fairly well simulated the atmospheric transfer and the soil surface temperature.

The simulation using the measured pH was less satisfactory (Fig. 6b). Fluxes were overestimated during DOY 78 to 79, owing to the surprisingly high values of the measured pH. But the diurnal and the hourly pattern were well described. This pointed out the problem of measuring pH in the field. Soil surface pH greatly changes when slurry is applied to a soil, due to the differing buffering capacities of the soil and the slurry. The procedure used to determine pH was based on the standard procedure. But this

type of measurement cannot give a value of the actual pH at the soil surface. It is rather a repeatable procedure that can be used to compare soils or to estimate a change of pH over time. It greatly modified the soil–slurry conditions by diluting and mixing. It would be better to use a portable pH-meter and a probe suitable for measuring pH directly on the soil surface or within the soil–slurry complex. This is an important problem that must be solved for any further development of ammonia volatilization models.

4.2. Adequacy of data and parameters used

Comparison with the values obtained for a loam soil belonging to the fourth soil type of the Clapp and Hornberger (1978) classification showed that the estimated porosity was slightly smaller than the measured one. But the sensitivity analysis showed that its effect was small. It has also been shown that the simulated infiltration was faster than the observed, despite this lower porosity. And the soil water content at field capacity was significantly underestimated by the pedo-transfer functions. Simulations with varying soil water contents before slurry spreading showed that the slurry infiltration was even more rapid for a rather dry soil, and that the uppermost soil layer dried too quickly, if b_W was underestimated. This indicates that it might be important to determine the specific b_W or soil water content at field capacity under the conditions at which the model is to be used, especially when spreading slurry over dry soils. But slurry is most often spread in autumn or early spring, when the soil water content is at field capacity. This limitation would not alter the operationality of the model. Another limitation of the model is that it does not simulate slurry ponding at the soil surface. The water and aqueous ammoniacal N of the slurry infiltrated too fast. Consequently, ammoniacal N availability at the surface decreased too fast, and ammonia fluxes were underestimated, especially for the first day. It was in fact difficult to quantify the effect of the structural degradation of the soil surface layers on water infiltration. One problem is that the water supplied with the slurry is added to the uppermost soil layer in the model. But, in fact, the aqueous phase of the slurry remains above the soil surface for several hours before actually entering the soil. The model might be improved by adding another layer to the model, corresponding to the slurry above the soil surface. The depth of this layer should decrease as slurry infiltrates and evaporates, and increase as it rains.

The present model cannot be considered to be fully operational, because it does not account for the ammonium adsorbed by the soil or for the soil–slurry pH change. The difficulties mentioned earlier suggest that no mechanistic model describing these processes is likely to be available in the near future. It would be therefore useful to have empirical relationships between the chemical properties of soil and slurry and their adsorption and buffering capacities.

4.3. Comparison with other models of ammonia volatilization

Comparison with empirical models shows the wide range of applications for this model. Empirical models analyze the influence of one or two factors related to the soil, agricultural practices or climatic conditions on volatilization. Consequently, they can only be used to predict loss under conditions similar to those for which they were established. Fluxes are generally given as a function of averages over the whole volatilization period. Consequently, the empirical models cannot account for the large variations in ammonia emissions in the first hours, which could explain much of the variability in ammonia losses through volatilization. They also cannot give any indication of the hourly change in fluxes, which would be useful for modelling the atmospheric chemistry.

Comparison with both empirical and mechanistics models indicates the possible practical use of this model. One of our objectives was to build a model that is easy to use, with no specific or labour demanding measurements. This has been achieved by using submodels constructed by other researchers with the same practical objective. Thus, the energy balance model used to calculate the soil surface temperature and evaporation employs standard meteorological data. Similarly, the advection model can be used to calculate the ammonia fluxes for field areas from several m^2 to several hectares. These submodels are added to the main model to minimize and simplify the input data. There are more input

data and parameters than for the empirical models, but they are not more difficult to obtain: meteorological data are now readily available at hourly intervals with automated weather stations, standard soil analysis gives the main soil characteristics, and the slurry parameters are obtained by standard analyses made before spreading. Consequently, this model can be easily used under real agricultural conditions for sensitivity analysis, experimental testing, predictive simulations, or for research applications. It differs in one major aspect from the other mechanistic models. It is the only mechanistic model that has been tested in agricultural conditions.

4.4. Using the model

4.4.1. Studying factors influencing ammonia volatilization

This model is suitable for studying the interaction of the chemical, physical and environmental factors on ammonia volatilization under field conditions. This can be illustrated by the following examples. The first concerns the influence of the meteorological conditions on ammonia losses by volatilization. Calculations showed that the likely changes in meteorological data, corresponding to slurry application on different days or seasons, could change the total losses by 10–20%. Fig. 7a shows that increasing the air temperature by 2 or 4°C increases ammonia losses by about 10 and 20%. Changing the wind speed alters the atmospheric transfer conditions. Multiplying the wind speed by 0.5 and 1.5 changes ammonia loss by −26 and +17%, respectively. Rainfall can reduce volatilization by decreasing the ammoniacal nitrogen concentration at the soil surface, and by reducing ammmonia gas transfer in the topsoil (Fig. 7c). A rainfall rate of 0.5 mm h^{-1} decreases the loss of ammonia by 20–25%

The second example examines the influence of the soil conditions at the time of slurry application. Using a water content at field capacity (value 1 on the X-axis) as a reference, Fig. 7d shows that decreasing the soil water content slightly increases volatilization, while increasing the soil water content

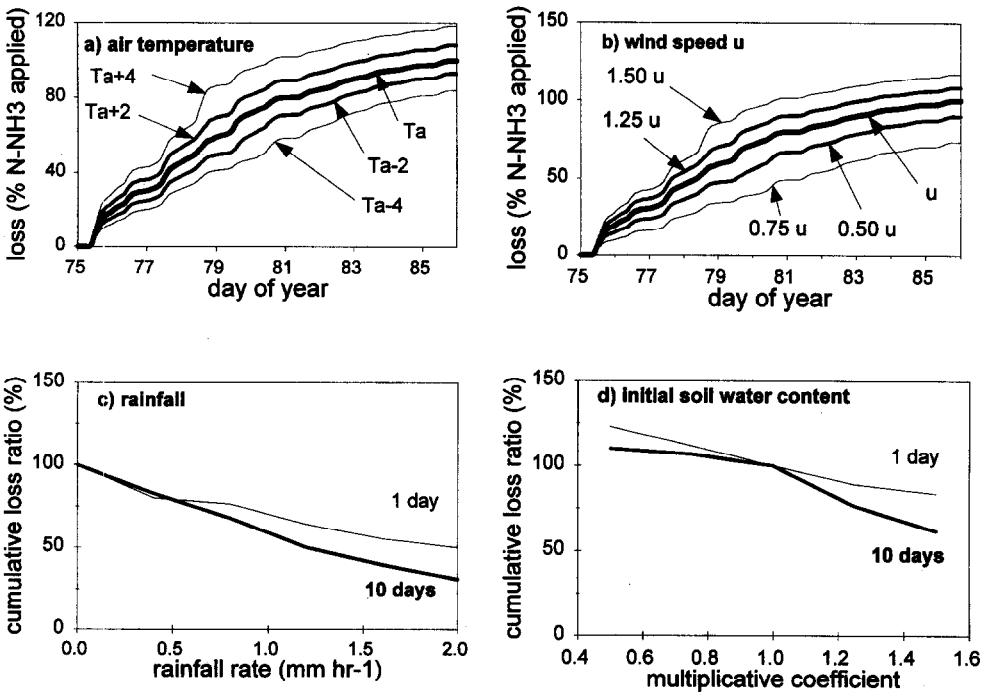


Fig. 7. Influence of climate and soil conditions on ammonia emissions. (a) air temperature; (b) wind speed at 2 m; (c) rainfall; (d) initial soil water content.

Table 3
Techniques and strategies for reducing ammonia volatilization

Parameter	Value	Calculated ammonia loss	Reference	Value	Measured ammonia loss
pH: acidification	7.7	100%	Stevens et al., 1989	7.0	100%
	7.2	53%		6.0	19%
	6.5	20%		6.5	12%
				4.0	3%
ammoniacal N content: dilution	0	100%	Stevens et al., 1992	0	100%
with water (% by volume of slurry)	43	78%		40	83%
	100	54%		100	50%
				140	39%
irrigation after spreading	0 mm	100%		0 mm	100%
	5 mm	68%	Klarensbeek and Bruins, 1991	10 mm	32%
	10 mm	62%	Moal et al., 1995	9 mm	67%
	20 mm	52%			

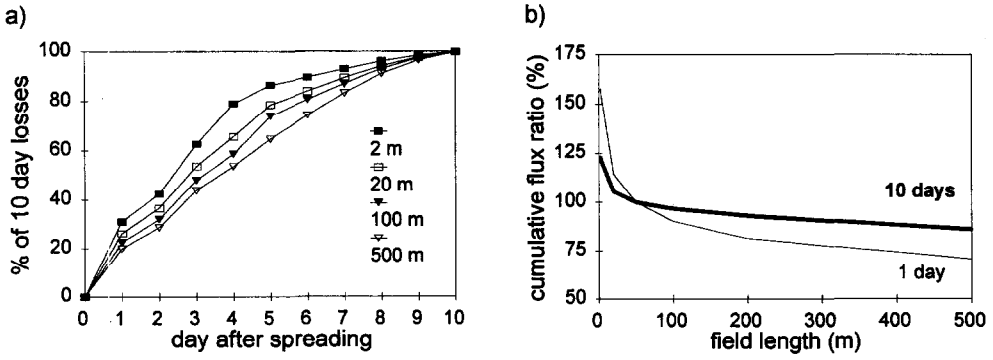


Fig. 8. Ammonia losses from experimental plots of various sizes.

more significantly decreases volatilization. This can be interpreted as a small effect on infiltration due to ponding of the slurry at the surface for a small decrease, and a larger effect by dilution due to a large increase in soil water.

The third example assesses the influence of agricultural practices on volatilization. Many experiments have been performed to estimate the effect of different techniques believed to reduce nitrogen losses by volatilization (see e.g., Stevens et al., 1989, 1992; Sommer and Husted, 1995a,b). However, their results are difficult to extrapolate to different conditions, because there are strong interactions with the climatic, soil and slurry conditions. A mechanistic model may help to explain the differences between experiments performed under different conditions, and to extrapolate the results, by integrating the factors related to climate, soil and slurry. Table 3 compares observed or calculated reductions in ammonia volatilization. A reduction in the pH by 1 unit or so gives a calculated reduction of 19% (for pH from 7.0 to 6.0) which fits fairly well the observed reduction (20% for a drop in pH from 7.7 to 6.5). Diluting slurry is a classical technique for reducing ammonia volatilization. The calculated reduction (54%) in volatilization obtained by diluting the slurry by 100% is close to the observed decrease (50%) measured by Stevens et al. (1992). And rainfall has a similar influence on ammonia volatilization as measured by Moal et al. (1995).

4.4.2. Influence of the measurement methods on emissions

Many results have been obtained with wind-tunnels, using an experimental area of 1 m^2. Volatilization should be much larger over small plots at the beginning of the volatilization period because of local advection, and it should diminish sooner because less ammonia is then available. Consequently, interpretation of the results must account for the size of the plots and the measurement methods used. This effect is quantified in Fig. 8, in which the normalized calculated ammonia loss are plotted against the size of the experimental plot. It confirms that the size of the plot changes the total amount of volatilized ammonia and the dynamics of the loss. The cumulative losses gradually increase over the large plots (100 m and 500 m), while they are very low after the fifth day for the smaller plot. These observations confirm the assumptions used by Génermont et al. (1997) to explain why the loss during the first day after slurry application was much less in their experiment under real field conditions than those, often quoted, obtained over small plots.

5. Conclusion

This model provides a realistic picture of the basic processes involved in the volatilization of ammonia from slurry applied in the field. It accounts for all the equilibria and transfers which might influence the amount of ammonia volatilized. The objective of producing an operational model has almost been realized. The main points remaining concern ammonium adsorption by the soil and the simulation of changes in pH. These require better estimations for the submodel of physical and chemical equilibria. Slurry infiltration also requires the addition of a

layer of slurry, and this will require better knowledge of the capacity of the slurry to retain water. The reduced infiltrability due to structural degradation of the soil surface after slurry spreading will also have to be better parameterized: it is known to be influenced by the soil constitution and its initial water content (Fiès and Panini, 1995). Despite all these possible improvements, it is possible to calibrate this model with satisfactory accuracy, so that it gives a fair picture of the change in ammonia emissions with time at hourly and daily time intervals, and these changes compare well with measured values in the field.

As this model might be applied to a wide range of climatic, soil and slurry conditions, it is designed to act as a base for predicting ammonia volatilization. Consequently, it should also be tested under a wide range of agricultural, pedological and climatic conditions in real field conditions. The model must also be checked to ensure that it responds correctly to a change in external conditions. But, it is difficult and expensive to conduct experiments in real field conditions and to control environmental conditions. This could rather be achieved using wind-tunnels (Lockyer, 1984), where controlling soil and environmental conditions is relatively easy. This is possible because the model can account for the size of the plot.

One important aim of this kind of model is to link it to models simulating the dynamics of the nitrogen in the soil–plant system. This means taking into account nitrogen transformations in the soil: organic N mineralization, ammmoniacal N organization, oxidation or nitrification. This will provide better management of nitrogen fertilizers, animal manures and sludges, because the environmental impacts of some agricultural practices on aquifer pollution by nitrates, acidification and global warming must be reduced. Most of these models work with a daily time scale, and account very roughly for nitrogen losses by volatilization. As our model uses input data compatible with these models, it could easily be used to calculate simple relationships that provide more realistic estimates of volatilization losses on a daily time scale, as a function of soil, agricultural and meteorological factors.

Lastly, this model may be used to improve the emission compartment of deposition models. These models—which integrate emission, dispersion, chemical transformation and deposition submodels —were first developed by researchers involved in atmospheric research. But as there was no suitable volatilization models, they did not account for the temporal variation in emissions at hourly intervals, which is the basic time scale of these models. Its mechanistic description of the processes involved in ammonia volatilization, and the availability of its input data, make our model suitable for integration into these deposition models. This would certainly improve the emission compartment of these deposition models by accounting for the temporal variations in fluxes using a physically-based approach.

6. Symbols

Symbol	Significance	Equation	Unit
a	soil albedo	Eq. (15)	dimensionless
a_i, b_i	coefficients of the advection model	Eq. (12)	
a_s, b_s	coefficients of the adsorption Freundlich isotherm	Eq. (7)	
b_W	coefficient of the pedo-transfer functions	Table 1	dimensionless
C	volumetric heat capacity	Eq. (9)	$J\ m^{-3}\ K^{-1}$
c_p	air specific heat at constant pressure	Eq. (19)	$J\ kg^{-1}\ K^{-1}$
$D^{W}_{NH_4^+}$	aqueous ammonium ions diffusivity in soil solution	Eq. (10)	$m^2\ s^{-1}$
$D^{aq}_{NH_4^+}$	aqueous ammonium ions diffusivity in pure water	Table 1	$m^2\ s^{-1}$
$D^{g}_{NH_3}$	ammonia diffusivity in soil atmosphere	Eq. (11)	$m^2\ s^{-1}$
$D^{air}_{NH_3}$	ammonia diffusivity in free air	Table 1	$m^2\ s^{-1}$
e_a	atmospheric vapour pressure	Eq. (18)	hPa

E	soil evaporation	Eq. (14)	W m^{-2}
F	Faraday constant (= 96484.56)	Table 1	C
G	soil heat flux	Eq. (14)	W m^{-2}
h	relative humidity at soil surface	Eq. (20)	dimensionless
h_W	soil water pressure head	Eq. (8)	m
h_{Wsat}	soil water pressure head at saturation	Table 1	m
H	atmospheric sensible heat flux	Eq. (14)	W m^{-2}
I	ionic force		mol l^{-1}
J_W	vertical flow of water in the soil	Eq. (8)	m s^{-1}
k	von Karman constant (= 0.40)	Eq. (12)	dimensionless
K_A	acid base equilibrium constant	Eq. (3)	dimensionless
K_H	Henry's law constant	Eq. (4)	atm l mol^{-1}
K_W	hydraulic conductivity of the soil	Eq. (8)	m s^{-1}
K_{Wsat}	hydraulic conductivity of the soil at saturation	Table 1	m s^{-1}
n	cloudiness	Eq. (17)	dimensionless
P_{NH_3}	ammonia partial pressure	Eq. (4)	atm
r_a	soil surface aerodynamic resistance	Eq. (19)	s m^{-1}
R	gas constant (= 0.831)	Table 1	atm l mol^{-1} K^{-1}
R_a	atmospheric radiation	Eq. (15)	W m^{-2}
R_n	soil surface net radiation	Eq. (14)	W m^{-2}
R_s	solar radiation	Eq. (15)	W m^{-2}
t	time	Eq. (10)	s
T	temperature	Eq. (9)	K
T_a	air temperature	Eq. (19)	°C
T_{ad}	daily average of air temperature	Eq. (16)	°C
T_d	dew point	Eq. (20)	°C
T_s	soil surface temperature	Eq. (15)	°C
u_d	daily average of wind speed	Eq. (22)	m s^{-1}
u^*	friction velocity	Eq. (12)	m s^{-1}
x	abscissa in wind direction	Eq. (12)	m
X	field length	Eq. (13)	m
z	height or depth		m
z_0	roughness length for the wind speed profile	Eq. (12)	m
$Z_{NH_4^+}$	ionic charge of ammonium ions	Table 1	C
ε_a	atmospheric emissivity	Eq. (16)	dimensionless
ε_{aB}	clear sky atmospheric emissivity	Eq. (17)	dimensionless
ε_s	soil emissivity	Eq. (15)	dimensionless
θ_W	soil water content	Eq. (8)	m^3 m^{-3}
θ_{Wfc}	soil water content at field capacity	Eq. (21)	m^3 m^{-3}
θ_{Wsat}	soil water content at saturation	Table 1	m^3 m^{-3}
θ_{Wsurf}	soil surface water content	Eq. (21)	m^3 m^{-3}
γ	psychrometric constant	Eq. (20)	Pa K^{-1}
$\gamma_{NH_4^+}$	crystal ionic radius of ammonium ion	Table 1	m
λ	latent heat of vaporisation	Eq. (14)	kg m^{-2} s^{-1}
$\lambda_{NH_4^+}$	limiting equivalent conductance of ammonium ions	Table 1	
λ_S	soil thermal conductivity	Eq. (9)	W m^{-1} K^{-1}
ρ	air density	Eq. (19)	kg m^{-3}
ρ_{NH_3}	atmospheric ammonia density	Eq. (12)	μg m^{-3}

ϕ	soil porosity	Eq. (11)	$m^3\ m^{-3}$
σ	Stefan Boltzman constant ($\sigma = 5.67\ 10^{-8}$)	Eq. (15)	$W\ m^{-2}\ K^{-4}$
φ_{GH}	phase shift function in Eq. (22)	Eq. (22)	dimensionless
Φ_{NH_3}	ammonia gas flux	Eq. (12)	$kg\ m^{-2}\ s^{-1}$

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